

healthcare, autonomous vehicles, industrial automation, and not just voice and data. 5G networks are modular, software-defined, and scalable. They can slice bandwidth, prioritise latency-sensitive applications, and deliver customised services for different users.

The problem? They are still limited by geography. Towers cannot cover oceans or high mountains. SATCOM faces the opposite challenge: limitless reach but historically limited capacity. The logical step is to combine both—leveraging the strengths of each.

That is precisely what the 3rd generation partnership project (3GPP), the global body governing mobile standards, has done. With Release 17, it formally introduced non-terrestrial networks (NTN), a standardised framework for integrating satellite systems with terrestrial 5G networks. This means a 5G device could one day switch effortlessly from a cell tower to a satellite link, without the user even noticing.

Inside the architecture: How the hybrid network works

In simple terms, the integration happens at two levels: direct access and indirect access (backhaul).

• **Direct-to-satellite access.**

Devices (phones, IoT terminals, or gateways) equipped with satellite-compatible radios connect directly to orbiting satellites. The satellites then route traffic to the ground station and onward to the 5G core network. This model is ideal for remote or mobile users, such as aircraft, ships, or emergency responders, who operate outside terrestrial coverage zones.

• **Satellite backhaul (indirect access).** Standard 5G phones connect to local towers or base stations (gNodeBs). Those stations, instead of relying on fibre, use satellite links for backhaul connectivity. This is how Vodafone recently made a 5G satellite call using a regular

From the ground to the sky: Real-world examples

From skies to seas and remote industries, the fusion of 5G and satellite networks is unlocking uninterrupted connectivity everywhere. This powerful partnership extends reliable, high-speed communication to places once beyond reach—transforming how people travel, work, and respond to emergencies.

- **Starlink's India debut.** Maharashtra becomes the first state to partner with Elon Musk's SATCOM firm.
- **Aviation connectivity.** Air India's new in-flight Wi-Fi uses LEO satellites for true internet access. Aircraft antennas connect to satellites, which relay data to ground gateways, delivering broadband at 10,668 metres (35,000 feet).
- **Maritime communication.** Ships now use compact LEO terminals instead of costly geostationary links, ensuring constant broadband for navigation, safety, and crew welfare.
- **Industrial IoT and smart mines.** Portable 5G gateways with satellite links keep factories and mobile mines connected, enabling seamless machine-to-machine communication.
- **Disaster recovery and events.** When terrestrial networks fail, drones or high-altitude balloons linked to satellites restore emergency connectivity instantly.
- **Connected mobility.** Satellite-backed 5G enables vehicle-to-vehicle and infrastructure communication beyond cities, powering safer, smarter autonomous travel.

As terrestrial and satellite systems converge, connectivity becomes truly universal. Whether in the air, at sea, underground, or on the move, this new era of 5G-SATCOM integration ensures that no location is ever out of reach.

smartphone; the phone communicated with a tower, and the tower relayed data via satellite.

Both models rely on gateways, modems, and software interfaces standardised by 3GPP, such as N6, N3, and NR-U interfaces, to ensure interoperability between terrestrial and space-based nodes.

Technically, when a 5G network collaborates with satellites, it becomes part of what's known as an NTN (non-terrestrial network) ecosystem. The traditional ground-based radio unit is replaced or supplemented by a satellite-capable gateway, while the 5G core remains unchanged—showcasing how flexible and future-ready 5G truly is.

The invisible backbone: Spectrum, standards, and interfaces

The secret behind seamless communication lies in harmonised spectrum and standardised interfaces. The 3rd generation partnership project (3GPP) has earmarked specific frequency bands for satellite operations under the NTN framework. These range from traditional Ku and Ka bands to newer S and X bands

optimised for low-latency links.

For engineers, the critical interfaces include:

- N6 – Connecting the 5G core to external data networks
- N3 – Linking the base station (gNodeB) to the 5G core
- NR-UE (new radio user equipment) – Devices capable of communicating with non-terrestrial radio links

Such modularity means the 5G core does not require re-engineering to work with satellites. Only the radio access layer changes, replaced by satellite-capable gateways or transceivers. This approach ensures that both existing infrastructure and future upgrades remain interoperable—a crucial factor for commercial deployment.

The economics: Falling costs, rising opportunity

Historically, launching and maintaining satellites was prohibitively expensive. However, in the past two decades, the cost of putting a kilogram into orbit has dropped by over 90 per cent. Reusable launch vehicles, miniaturised electronics, and mass production have